SECSR2043 OPERATING SYSTEMS
[20 Marks]

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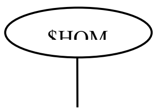
Instruction: Please answer all of the following questions. Whenever the ☝ symbol appears, please raise your hand to call your instructor, he/she will verify your results by putting his / her initial next to the symbol.

1. Type the following commands using a text editor and save it as a *yourname*.sh (Example: `ahmad.sh`).

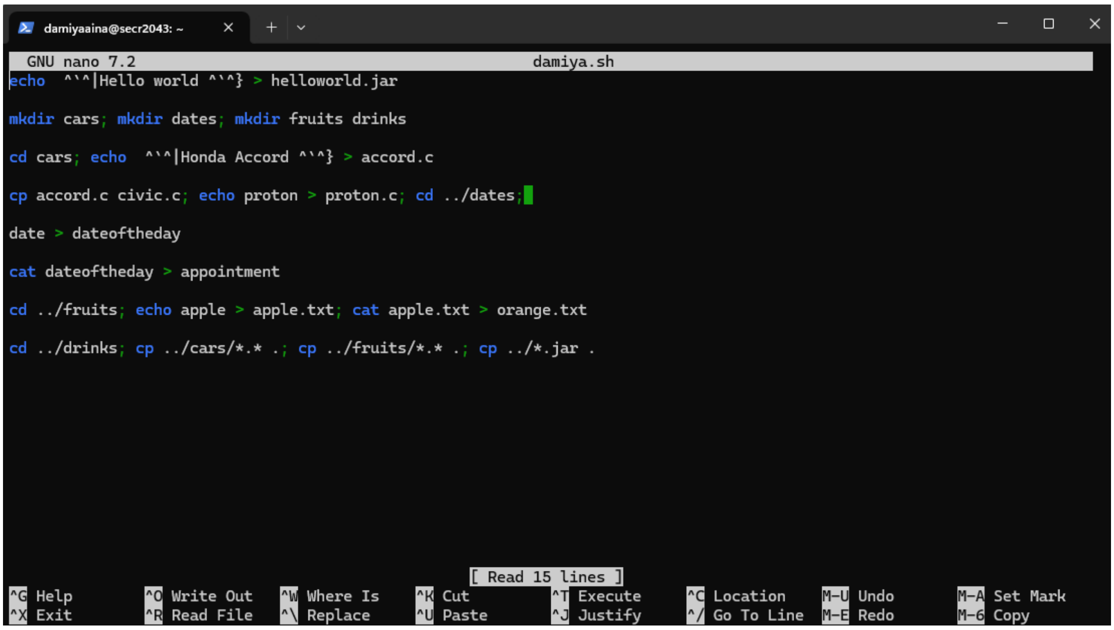
```
echo "Hello world" > helloworld.jar
mkdir cars; mkdir dates; mkdir fruits drinks
cd cars; echo "Honda Accord" > accord.c
cp accord.c civic.c; echo proton > proton.c; cd ../dates;
date > dateoftheday
cat dateoftheday > appointment
cd ../fruits; echo apple > apple.txt; cat apple.txt >
orange.txt
cd drinks; cp ../cars/*.* .; cp ../fruits/*.* .;
cp ../*.jar .
```

a) Execute the script and draw a tree structure that contains created directories and files. The parent node of the directory begin with **$HOME** directory.

[4 marks]

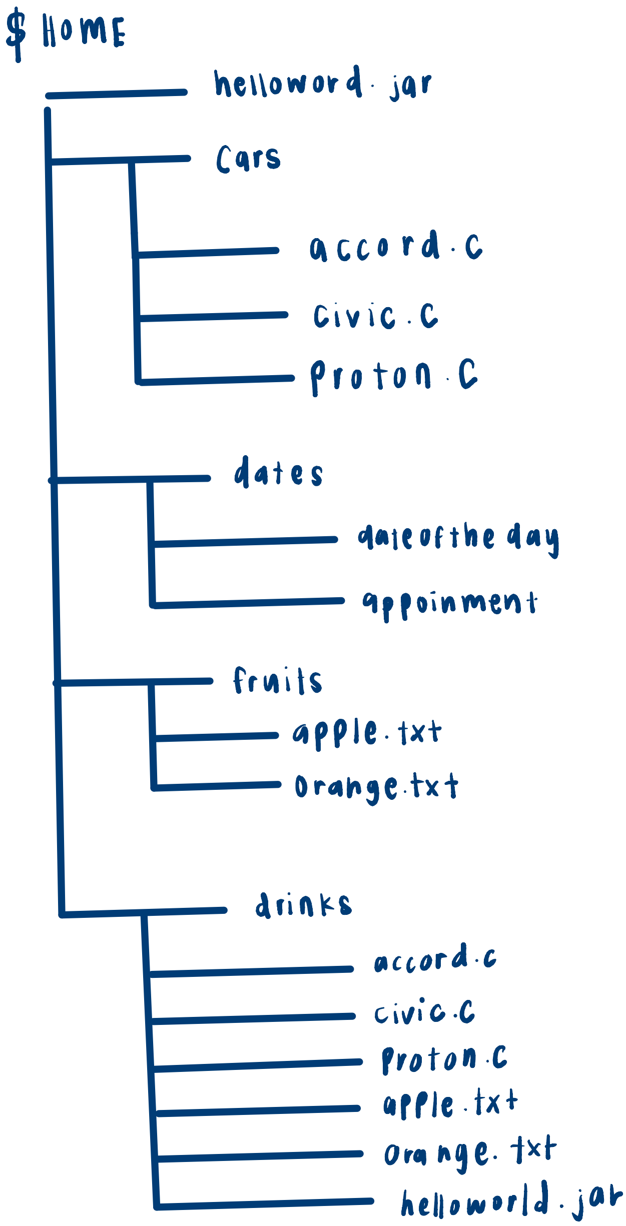


Print screen the script that you type;



```
damiyaaina@secr2043: ~
GNU nano 7.2                                   damiya.sh
echo  ^`^|Hello world ^`^} > helloworld.jar

mkdir cars; mkdir dates; mkdir fruits drinks

cd cars; echo  ^`^|Honda Accord ^`^} > accord.c

cp accord.c civic.c; echo proton > proton.c; cd ../dates;

date > dateoftheday

cat dateoftheday > appointment

cd ../fruits; echo apple > apple.txt; cat apple.txt > orange.txt

cd ../drinks; cp ../cars/*.* .; cp ../fruits/*.* .; cp ../*.jar .

                                [ Read 15 lines ]
^G Help      ^O Write Out  ^W Where Is   ^K Cut     ^T Execute   ^C Location    M-U Undo   M-A Set Mark
^X Exit      ^R Read File  ^\ Replace    ^U Paste   ^J Justify   ^/ Go To Line  M-E Redo   M-6 Copy
```

Then draw the tree



b) Write an interactive bash script that will read a type of file extension, display all those files, and count the number of files. To validate your script, display `c` program files, and enter "`c`" as the input to the bash script. [4 marks]

Print screen the bash script you type and run

```
damiyaaina@secr2043: ~
GNU nano 7.2                                    damiya2.sh
#!/bin/bash

echo "Enter file extension (without dot): "
read ext

files=$(ls *.$ext 2>/dev/null)

if [ -z "$files" ]; then
    echo "No files with .$ext extension found."
    echo "Total files: 0"
else
    echo "Files with extension .$ext:"
    echo "$files"
    echo "Total files: $(echo "$files" | wc -l)"
fi
```

```
damiyaaina@secr2043:~$ nano damiya2.sh
damiyaaina@secr2043:~$ chmod +x damiya2.sh
damiyaaina@secr2043:~$ echo "int main() {}" > test1.c
echo "void fun() {}" > test2.c
damiyaaina@secr2043:~$ ./damiya2.sh
Enter file extension (without dot):
c
Files with extension .c:
test1.c
```

2. The following Figure 1 illustrates a tree structure of some directories and files.

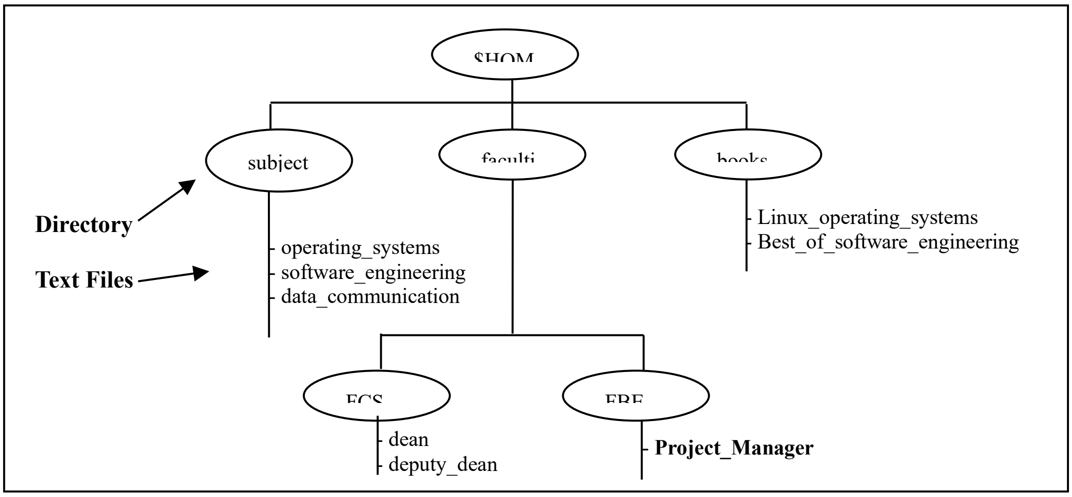


Figure 1

a) Write a bash script (called `myname2a.sh`) that will produce directories and files as in Figure 1. Each text files contain its filename without the underscore character. For example: text file `Project_Manager` contains `Project Manager`). [4 marks]

Print screen the bash script you type and run

```
GNU nano 7.2                                    damiya2a.sh
#!/bin/bash

# Create the directory structure
mkdir -p subjects/books/{operating_systems,software_engineering,data_communication}
mkdir -p faculties/FCS/{Linux_operating_systems,Best_of_software_engineering}
mkdir -p faculties/FBE/{dean,deputy_dean,Project_Manager}

# Create files with content inside
echo "Linux operating systems" > faculties/FCS/Linux_operating_systems/Linux_operating_systems
echo "Best of software engineering" > faculties/FCS/Best_of_software_engineering/Best_of_software_engineering
echo "Project Manager" > faculties/FBE/Project_Manager/Project_Manager
```

```
.
|-- cars
|   |-- accord.c
|   |-- civic.c
|   `-- proton.c
|-- damiya.sh
|-- damiya2.sh
|-- damiya2a.sh
|-- dates
|   |-- appointment
|   `-- dateoftheday
|-- drinks
|   |-- accord.c
|   |-- apple.txt
|   |-- civic.c
|   |-- helloworld.jar
|   |-- orange.txt
|   `-- proton.c
|-- faculties
|   |-- FBE
|   |   |-- Project_Manager
|   |   |   `-- Project_Manager
|   |   |-- dean
|   |   `-- deputy_dean
|   `-- FCS
|       |-- Best_of_software_engineering
|       |   `-- Best_of_software_engineering
|       `-- Linux_operating_systems
|           `-- Linux_operating_systems
|-- fruits
|   |-- apple.txt
|   `-- orange.txt
|-- helloworld.jar
|-- subjects
|   `-- books
```

```
|       |-- data_communication
|       |-- operating_systems
|       `-- software_engineering
|-- test1.c
`-- test2.c

18 directories, 22 files
```

b) Complete the following table by writing the access control of directories or files that were produced. Given is the access control for directory called `book`.

[2 marks]

Directory/File	Access Control
`books`	`drwxrwxr-x`
`subjects`	`dr-x--x--x`
`Best_of_software_engineering`	`drwxrwxr-x`
`FCS`	`drwxrwxr-x`
`project_manager`	`drwxr-xr-x`

c) Write another bash script (called `myname2c.sh`) that will change the access control of the directories and files based on the following information:

[4 marks]

Directory/File	Users								
	Owner			Group			Public		
`subjects`	✓	✓	✓	✓	x	x	✓	x	x
`Best_of_software_engineering`	✓	x	✓	x	✓	x	x	x	x
`FCS`	✓	✓	x	x	x	x	✓	✓	✓
`project_manager`	x	x	x	x	✓	✓	x	x	✓

Print screen the bash script you type and run

```
  GNU nano 7.2                                                   damiya2c.sh
#!/bin/bash

# Change permissions based on instructions

chmod 551 subjects
chmod 775 faculties/FCS/Best_of_software_engineering
chmod 775 faculties/FCS
chmod 755 faculties/FBE/Project_Manager
```

```
drwxr-xr-x 2 damiyaaina damiyaaina 4096 Jun 22 12:05 faculties/FBE/Project_Manager
drwxrwxr-x 4 damiyaaina damiyaaina 4096 Jun 22 12:05 faculties/FCS
drwxrwxr-x 2 damiyaaina damiyaaina 4096 Jun 22 12:05 faculties/FCS/Best_of_software_engineering
dr-xr-x--x 3 damiyaaina damiyaaina 4096 Jun 22 12:05 subjects
```

d) Complete the following table by writing the access control for each directory or file after executing the bash script in question 2(c)). [2 marks]

Directory/File	Access Control
subjects	dr-xr-x—-x
Best_of_software_engineering	drwxrwxr-x
FCS	drwxrwxr-x
project_manager	drwxr-xr-x

End of Lab 3

**** All the Best for Final Exam ****